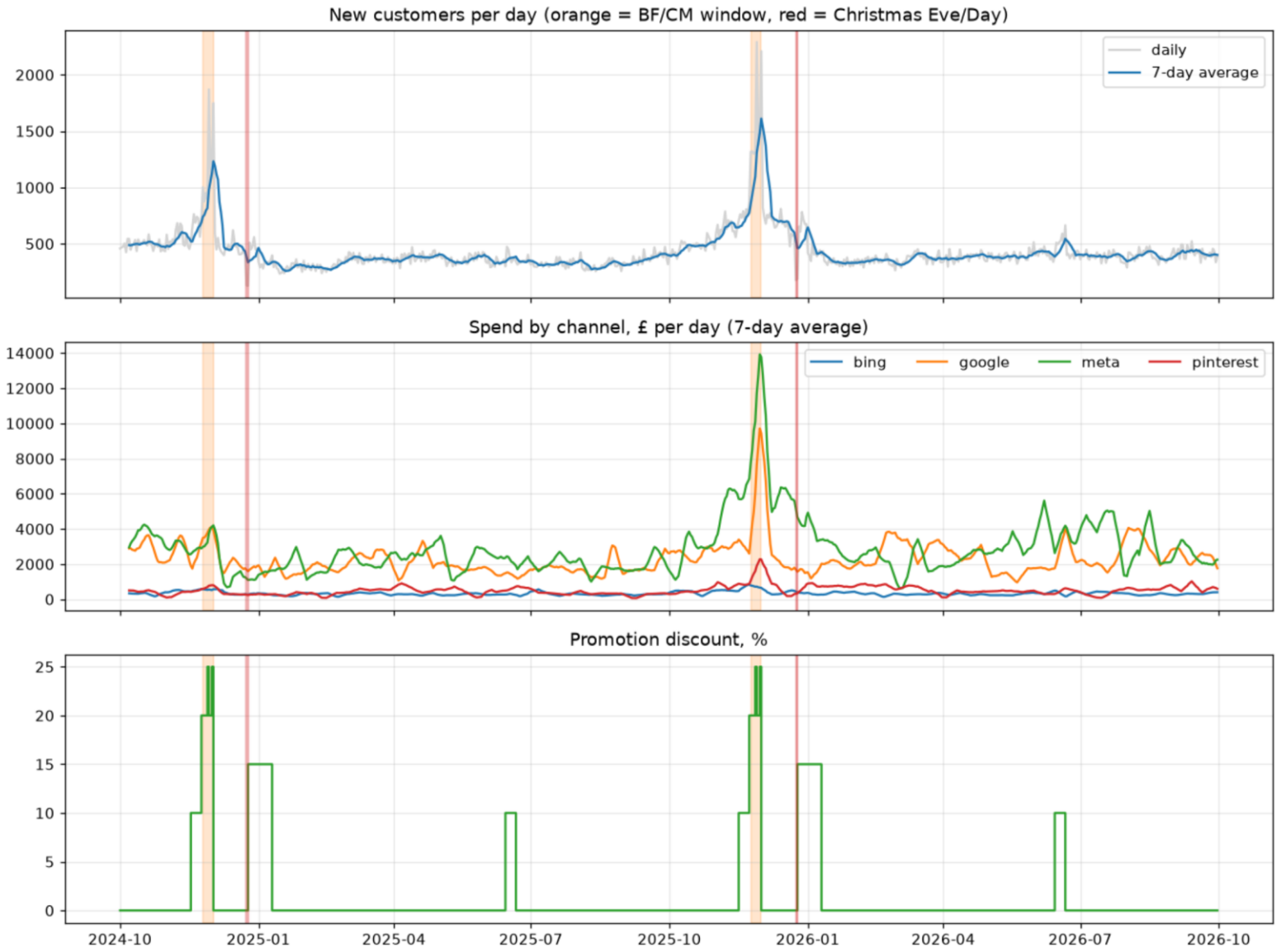
MARKETING MIX MODEL

Where to add budget to win new customers under £40

Marketing mix model, daily data October 2024 to September 2026

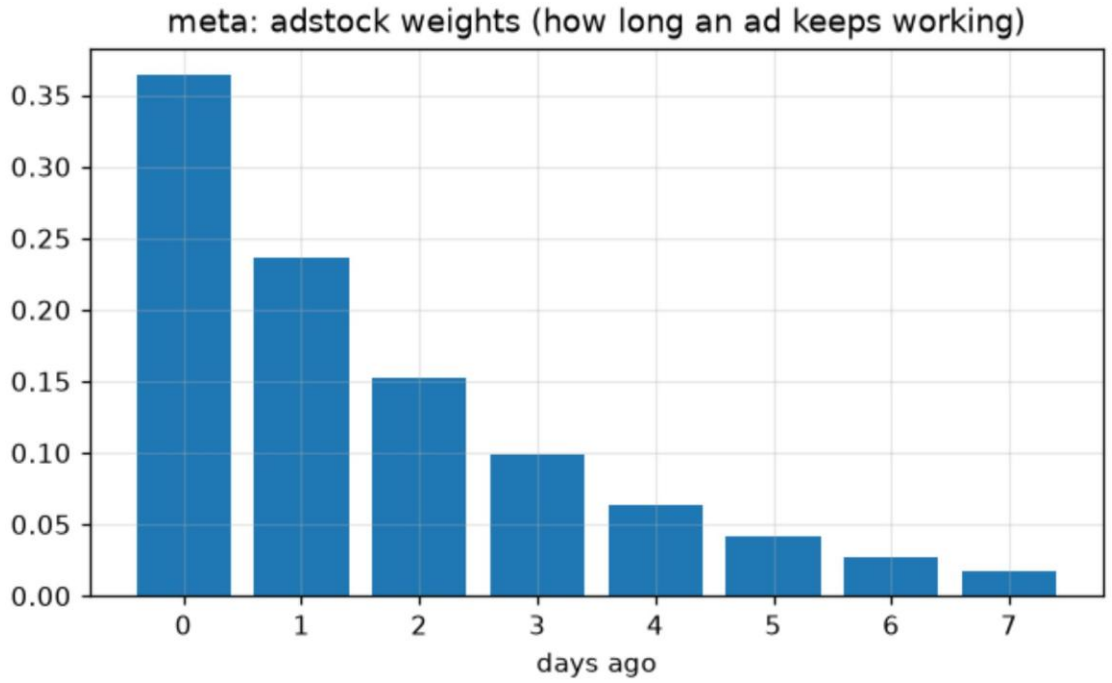
The question is which channels to scale, hold or reduce.

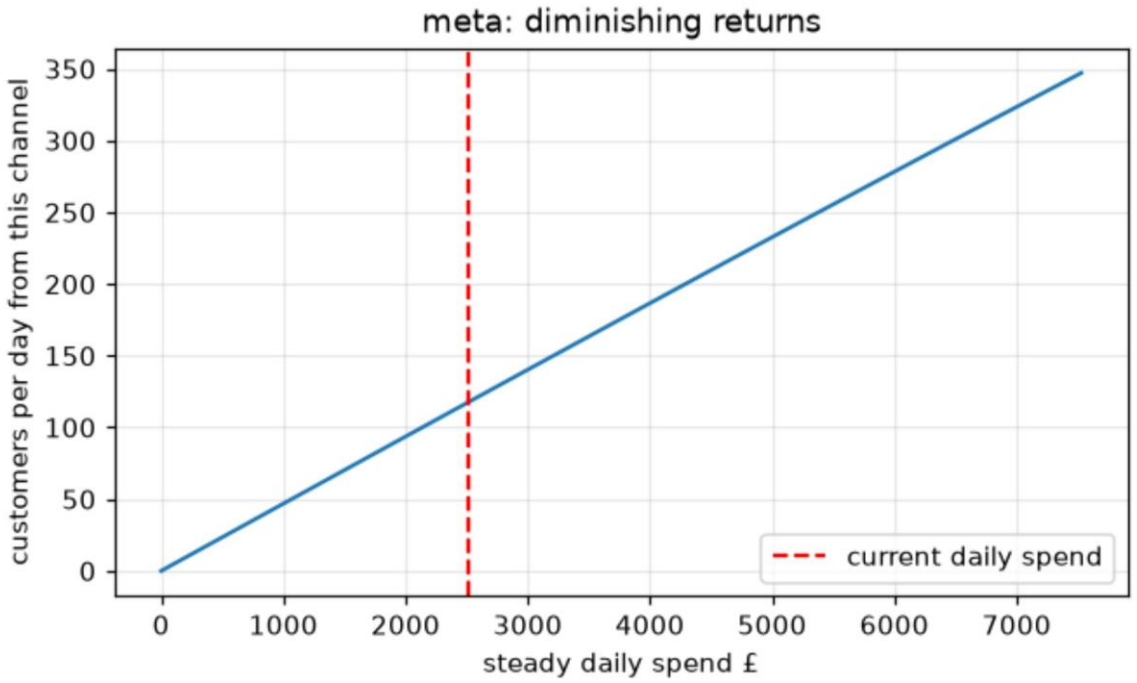


MODEL MECHANICS

How the model works

- Predicts daily new customers from spend in Bing, Google, Meta and Pinterest.
- Separates baseline, seasonality, promotions and Black Friday / Cyber Monday.
- Adstock: Meta’s ads keep working for days; half the effect is gone after about 1.6 days.
- Saturation: Meta’s response is almost a straight line up to 2.5× today’s spend.
- Each estimate is a range, not a single number.



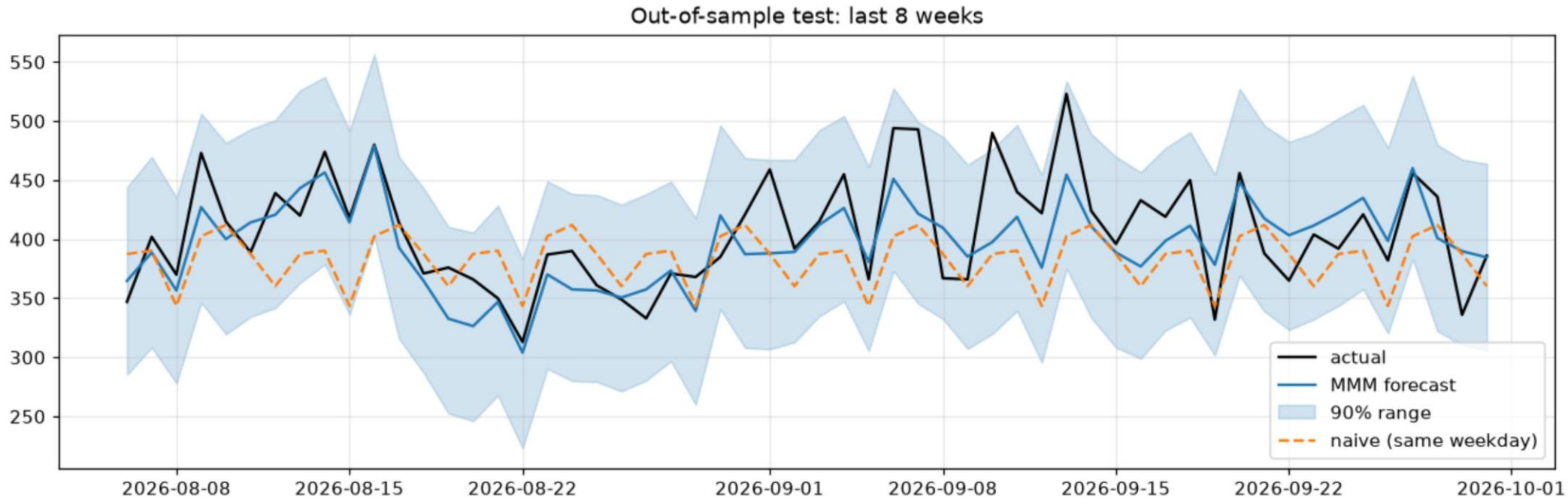


VALIDATION

Can we trust the model?

- Data checks: 10 of 10 passed. No gaps, duplicates or unit problems found.
- The sampler converged: max r-hat 1.005, no divergences.
- On the last 8 weeks held out from training, model average error 6.2%; simple forecasts scored 9.1–9.5%.
- The model beat the simple forecasts in all 8 weeks.
- Limit: ranges cover parameter uncertainty only, not the risk that the model structure is wrong.

Metric	Model	Same weekday, last 4 weeks	Flat 28-day average
MAE (customers/day)	25.7	38.6	40.4
MAPE	6.2%	9.1%	9.5%
RMSE	32.8	47.9	51.9

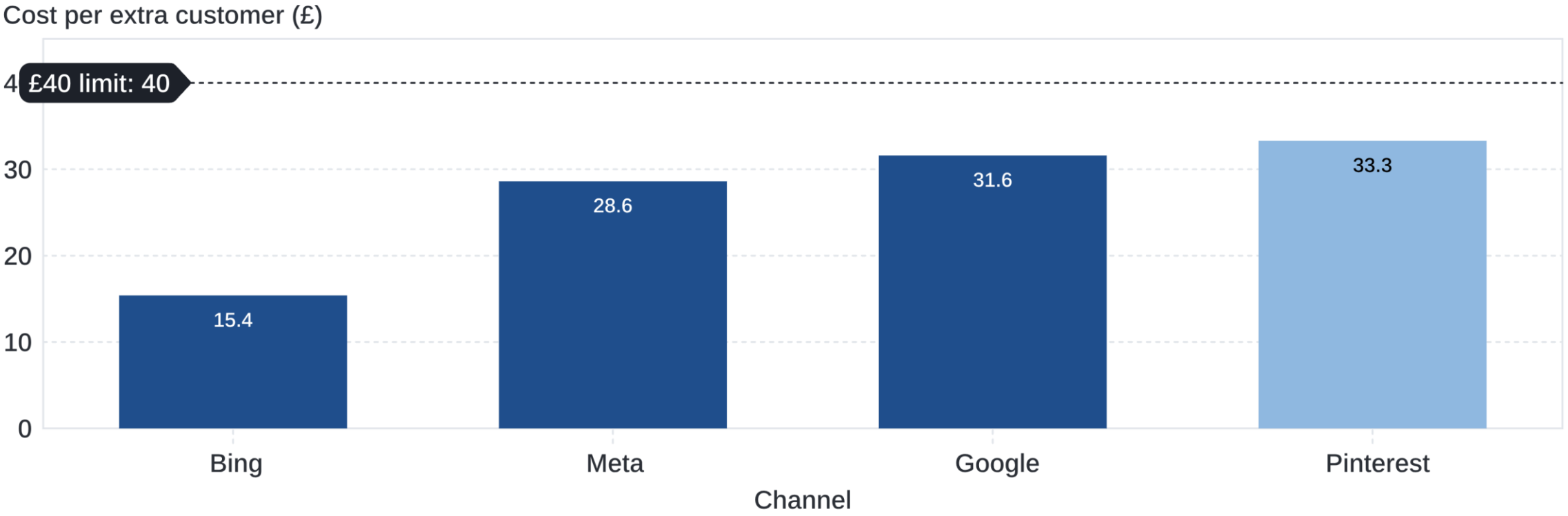


MARGINAL ECONOMICS

One extra customer costs £15–£33 today, all under £40

- Bing £15, Meta £29, Google £32, Pinterest £33 per extra customer (+10% step).
- Google reaches £40 at about 1.5× today's spend; Pinterest is least certain (71% chance of staying below £40).

Cost per extra customer, +10% step



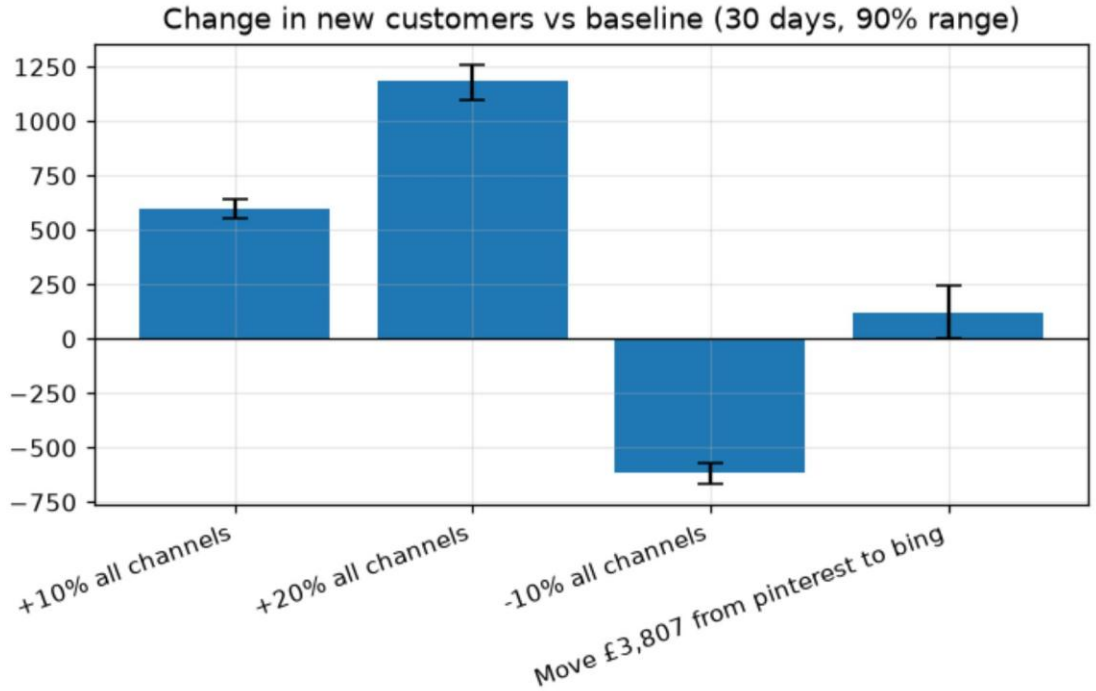
SCENARIO PLANNING

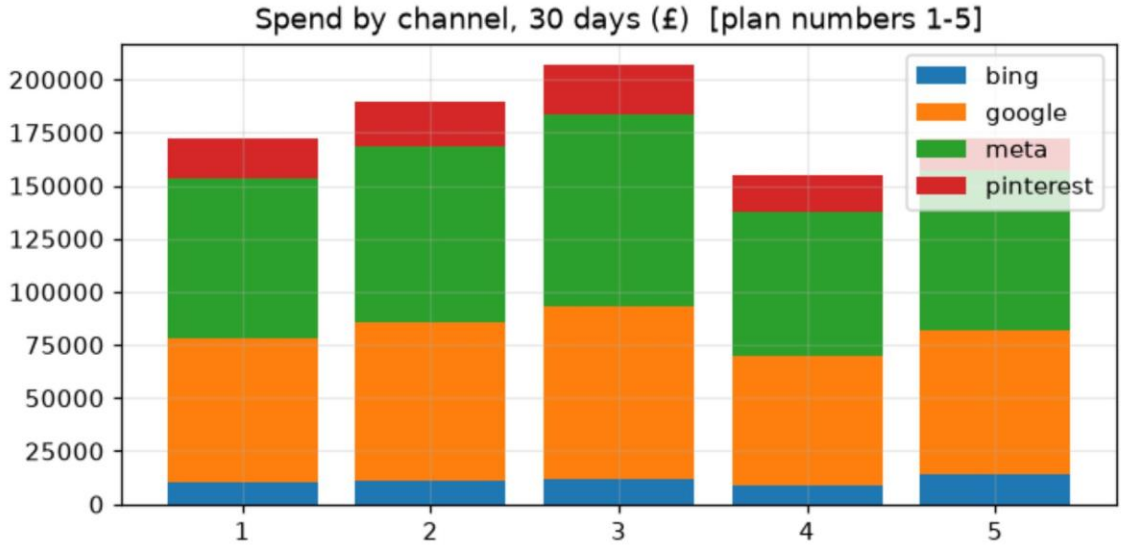
Adding 10–20% across channels brings about 600–1,200 more customers

Add budget to Meta first, then Bing; grow Google only to about 1.5× today's spend; hold Pinterest, and do not cut, since every customer costs under £40

Scenario (30 days)	Spend change	Extra new customers (90% range)	Cost per extra customer
Keep spend (baseline 12,558) (1)	£0	n/a	n/a
+10% all channels (2)	+£17.2k	+600 (557 to 641)	£28.7
+20% all channels (3)	+£34.4k	+1,183 (1,098 to 1,262)	£29.1
−10% all channels (4)	−£17.2k	−616 (−663 to −568)	£28 saved per customer lost
Move £3.8k Pinterest → Bing (5)	£0	+127 (0 to 245)	n/a

The reallocation range includes zero, so the gain is not certain.





Test a 20% Meta increase before scaling further

Acquiring a single customer costs Meta approximately £29, below the £40 limit, so it is the first place to test more budget.

Meta increase	Extra spend	Extra customers	Cost per extra customer
+10%	£7.5k	263	£28.6
+20% (test)	**£15.1k**	**526**	**£28.6**
+40%	£30.1k	1,050	£28.7
+80%	£60.2k	2,093	£28.8

Measure weekly with control regions kept at the old budget → review after 7 days → decide after 30 days

PLAN

- Test: Meta +20% for 30 days, £75.3k → £90.3k (+£15.1k)
- Expected: about 526 extra customers (range 488 to 566)
- Success: cost per extra customer at or below £40; reduce if above £40 for two weeks

LIMITS

- The range shows uncertainty in the estimate, not whether the model itself is right
- Only two Black Fridays in the data
- Correlation is not causation; the test will confirm the effect

Model estimate. The test will confirm or correct it.